

Graph-Structured Deep Learning for Financial Forecasting

A Hybrid THGNN–MaGNet Architecture and Multi-Agent Decision Pipeline for Indian Equity Markets (NSE)

Debarshi Chakraborty | ID B2430045

MSc Big Data Analytics | RKMVERI, Belur

Supervisor: Mr Champak Kumar Dutta | Dept. of Computer Science

1. Motivation & Problem
2. Background & Related Work
3. Contributions
4. Model Architecture
5. Training Methodology
6. Results
7. Multi-Agent Decision Pipeline
8. Conclusion

Motivation & Problem

Why Is Financial Forecasting Hard?

Stock prices are generated by **millions of interacting agents** who observe and anticipate each other.

Consequences for the data:

- Non-stationary — structural breaks, regime shifts
- **Very low signal-to-noise** — useful rank-IC is only 0.05–0.15
- Stocks are **not independent** — sector & macro shocks couple them
- Relationships are **dynamic** — correlations change over time

Two research gaps:

1. Most SOTA is tested only on US / Chinese markets — **NIFTY 500 underserved**

Task: Cross-Sectional Ranking

Input: $\mathbf{X}^\tau \in \mathbb{R}^{N \times T \times F}$

$N = 353$ NIFTY 500 stocks

$T = 20$ trading-day look-back

$F = 12$ OHLCV + technical

features

Output: $\hat{\mathbf{y}} \in \mathbb{R}^N$ — next-day return score

Objective: maximise rank correlation

$$\text{IC} = \text{corr}_{\text{Spearman}}(\hat{\mathbf{y}}, \mathbf{y})$$

Long top- K / short bottom- K : IC directly measures investment value.

Background & Related Work

The Forecasting Landscape — Two Lineages

Temporal modelling

- **Classical:** ARIMA, VAR, ARCH/GARCH — linear, interpretable, ✗ nonlinear dynamics
- **RNN:** LSTM, GRU — gated memory; ✗ **channel-independent** (stocks in isolation)
- **Transformer:** TFT (gated variable selection), iTransformer (variate tokens)
- **State-space:** Mamba (selective SSM, linear-time), SAMBA (Mamba + sliding-window attn)

This work fuses both lineages: a **MAGE temporal encoder** (RNN+SSM+MoE) with a **dual hypergraph** relational module.

Relational modelling

- **Static graphs:** GCN over supply-chain / ownership — handcrafted, cannot adapt
- **Dynamic hetero-graph:** **THGNN** (pos/neg correlation, daily), DHSTN
- **Hypergraphs:** STHAN-SR (learning-to-rank), MSTNN, Hermes (lead-lag)
- **SOTA hybrid:** **MaGNet** — Mamba + MoE + dual hypergraph

Survey Summary & Six Research Gaps

Method	Relational	M-Scale	Market
iTransformer	None	No	Global
Mamba	None	No	NLP
THGNN	Dyn. Hetero	No	US, CN
STHAN-SR	Hypergraph	No	US, JP
DHSTN	Dyn. Hypergraph	No	CN, US
MASTER	Stock Attn	No	CN, US
StockMixer	MLP Mixing	Yes	US
Hermes	Hypergraph	Yes	Multi
MaGNet	Dual Hyper	Partial	CN, US

All surveyed models target **US** / **Chinese** markets.

G1 **Joint structure–prediction** optimisation

G2 SSM–hypergraph integration (un-ablated)

G3 SSM + temporally-asymmetric lead-lag

G4 Market-regime-dynamic graph topology

G5 **Indian market (NIFTY 500)** applicability

G6 **System-level integration**
(GNN+news+risk+LLM)

This work addresses **Gaps 1, 2, 3, 5, 6**; Gap 4 and full ablation are left to future work.

Contributions

Three Inter-Linked Contributions

1. Baseline

NIFTY 500 THGNN

First publicly documented GNN forecasting result on the Indian NIFTY 500 universe (353 stocks).

rank-IC = 0.0204

2. Architecture

Hybrid THGNN–MaGNet

MAGE temporal encoder + **dual hypergraph** (TCH + GPH), trained jointly with a composite IC loss.

rank-IC = 0.0398 (+95%)

3. System

6-Agent Pipeline

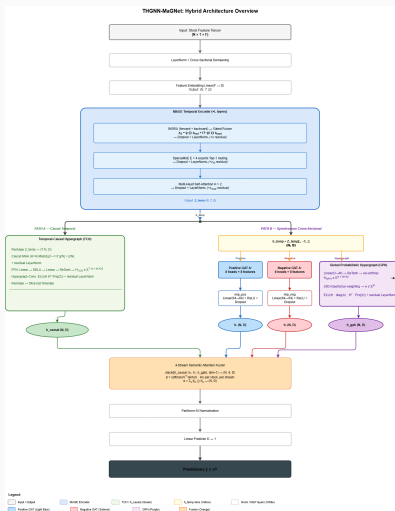
LangGraph routes GNN scores through FinBERT sentiment, risk, macro & LLM report agents.

Deployed via Streamlit

A single unified framework: **controlled baseline** → **novel hybrid model** → **deployable decision-support system**.

Model Architecture

THGNN–MaGNet — End-to-End Architecture

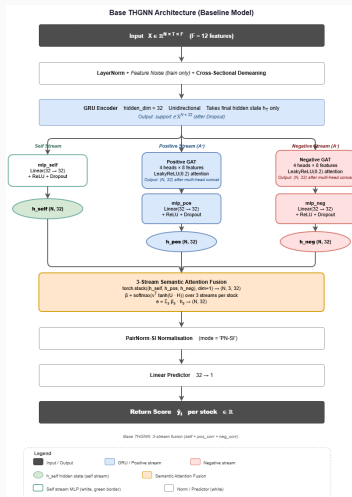


Shared **MAGE** temporal encoder,
then two parallel paths:

- **Path A — Causal Temporal:**
full sequence → Temporal-Causal Hypergraph (TCH)
- **Path B — Cross-Sectional:** final state → Pos-GAT, Neg-GAT, Global Hypergraph (GPH)

Four representations fused by **4-stream semantic attention** → PairNorm → linear head.

Base Model — THGNN Baseline



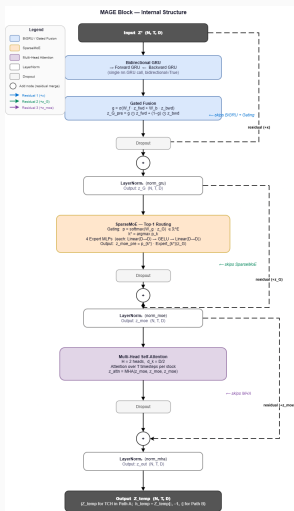
Re-implements THGNN on NIFTY 500 as the **controlled reference**.

- GRU temporal encoder (replaces original Transformer for the 6 GB GPU)
- Dynamic daily graph from rolling Pearson correlation
- Three streams**: self, positive-GAT, negative-GAT
- Semantic attention $\rightarrow$ PairNorm-SI $\rightarrow$ linear head

Adjacency from fixed thresholds:

$$A_{ij}^+ = \mathbf{1}[\rho_{ij} > \theta^+], \quad A_{ij}^- = \mathbf{1}[\rho_{ij} < \theta^-]$$

MAGE Temporal Encoder — Internal Structure



Three residual sub-layers, each with LayerNorm:

1. Bidirectional GRU + gated fusion

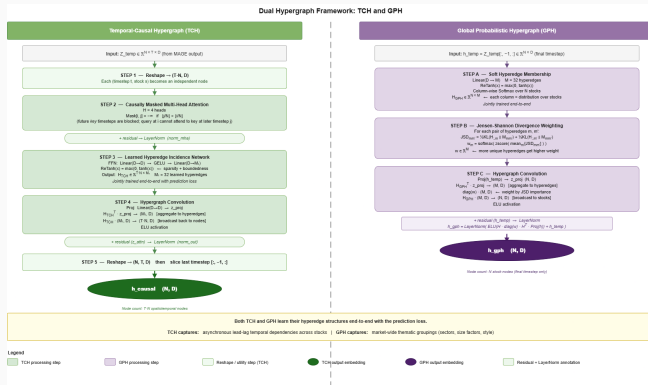
$$z^{\text{gru}} = g \odot z^{\text{fwd}} + (1 - g) \odot z^{\text{bwd}}$$

2. Sparse Mixture-of-Experts ($E=4$, top-1 routing) — experts specialise in trending vs. mean-reverting regimes, **without explicit regime labels**

3. Multi-head self-attention ($H=2$) contextualises across the T timesteps

Mamba variant swaps the BiGRU for a selective state-space scan.

Dual Hypergraph — Beyond Pairwise Edges



TCH (causal, lead-lag): causally-masked attention over $T \cdot N$ spatiotemporal nodes; learns $M_1=32$ hyperedges end-to-end.

$$\mathbf{H}_{\text{TCH}} = \text{ReTanh}(\mathbf{W}_{h2} \text{GELU}(\mathbf{W}_{h1} \mathbf{Z}_{\text{attn}}))$$

GPH (global themes): soft column-normalised membership ($M=32$ groups) with JSD-based importance weighting — captures latent sector-like clusters.

$$\mathbf{H}_{\text{GPH}} = \text{softmax}_{\text{col}}(\text{ReTanh}(\mathbf{W}_{\text{gph}} \mathbf{h}_{\text{temp}}))$$

TCH — Enforcing Temporal Causality (Gap 1 & 3)

The MAGE output is reshaped to $T \cdot N$ spatiotemporal nodes. **Causally-masked** multi-head attention prevents look-ahead leakage:

$$\mathbf{Z}_{\text{attn}} = \text{MHA}(\mathbf{Z}'; \text{Mask}_{ij} = -\infty \text{ if } \lfloor j/N \rfloor > \lfloor i/N \rfloor)$$

So node (t, s) *cannot* attend to any future node (t', s') , $t' > t$. **Hyperedges learned end-to-end** (the key novelty vs. MaGNet):

$$\mathbf{H}_{\text{TCH}} = \text{ReTanh}(\mathbf{W}_{h2} \text{GELU}(\mathbf{W}_{h1} \mathbf{Z}_{\text{attn}})) \in \mathbb{R}^{TN \times M_1}$$

$$\mathbf{Z}_{\text{conv}} = \text{ELU}(\mathbf{H}_{\text{TCH}}(\mathbf{H}_{\text{TCH}}^{\top} \mathbf{Z}' \mathbf{W}_{\text{proj}})) + \mathbf{Z}_{\text{attn}}$$

Why it matters

- Topology trained **jointly** with the prediction loss — not fixed thresholds (THGNN) or loss-independent (DHSTN)
- Causal mask captures **asynchronous lead-lag** — one stock leading another
- $\text{ReTanh}(x) = \max(0, \tanh x)$ gives sparse, bounded membership

Causal representation taken from the final timestep: $\mathbf{h}_{\text{causal}} = \mathbf{Z}_{\text{conv}}[:, T-1, :]$.

Cross-Sectional GAT Streams & Computational Cost

Heterogeneous pairwise attention over $\mathbf{h}_{\text{temp}}$

— two typed GATs ($H_{\text{gat}}=8$, $d_k=8$):

$$\alpha_{ij}^h = \frac{\exp(\text{LReLU}(\mathbf{a}_h^\top [\mathbf{W}_h \mathbf{h}_i \| \mathbf{W}_h \mathbf{h}_j]))}{\sum_{k \in \mathcal{N}_i} \exp(\text{LReLU}(\mathbf{a}_h^\top [\mathbf{W}_h \mathbf{h}_i \| \mathbf{W}_h \mathbf{h}_k]))}$$

- A^+ — stocks moving together
- A^- — stocks moving inversely

Pairwise edges give an explicit inductive bias
complementing the soft learned hyperedges.

Model	Temporal	Relational
THGNN	$O(NTd_hF)$	$O(N^2d_h)$
Hybrid BiGRU	$O(NTD^2)$	$O((TN)^2D)$ TCH
Mamba-MoE	$O(NTDS)$	same modules

The TCH causal attention is $O((TN)^2D)$: with $T=20$, $N=353$, the 7060×7060 matrix is ≈ 800 MB across 4 heads — **tractable on a 6 GB GPU**. Mamba reduces the encoder to near-linear $O(NTDS)$.

4-Stream Fusion & Prediction Head

Semantic attention fusion of four streams

$$\mathbf{H} = [\mathbf{h}_{\text{causal}}; \mathbf{h}_{+}; \mathbf{h}_{-}; \mathbf{h}_{\text{gph}}]:$$

$$\mathbf{w} = \mathbf{V}^{\top} \tanh(\mathbf{U}\mathbf{H}), \quad \beta = \text{softmax}(\mathbf{w})$$

$$\mathbf{e} = \sum_{s=1}^4 \beta_{\cdot,s} \odot \mathbf{H}_{\cdot,s,:}$$

Each stock *adaptively* weights its

- causal temporal context
- positive / negative correlation neighbourhood
- global thematic group

PairNorm-SI Prediction Head

$$\hat{\mathbf{e}}_i = \frac{\mathbf{e}_i - \bar{\mathbf{e}}}{\sqrt{\varepsilon + \|\mathbf{e}_i - \bar{\mathbf{e}}\|^2}}$$

Prevents over-smoothing after graph aggregation.

$$\hat{\mathbf{y}} = \hat{\mathbf{e}} \mathbf{W}_{\text{pred}} \in \mathbb{R}^N$$

The pairwise GAT edges provide an explicit inductive bias that **complements** the soft learned hyperedges.

Training Methodology

Composite Loss — Aligned with the IC Objective

Plain MSE does not optimise *ranking*. A three-term composite loss is used:

Total Loss

$$\mathcal{L} = \underbrace{\lambda_{\text{mse}} \mathcal{L}_{\text{mse}}}_{(1) \text{ anchoring}} + \underbrace{\bar{\lambda}_{\text{ic}}(e) \mathcal{L}_{\text{ic}}}_{(2) \text{ ranking}} + \underbrace{\lambda_{\text{disp}} \mathcal{L}_{\text{disp}}}_{(3) \text{ anti-collapse}}$$

(1) Normalised MSE

$\mathcal{L}_{\text{mse}} = \frac{1}{N\sigma_f^2} \sum_i (\hat{y}_i^{\text{cs}} - y_i^{\text{cs}})^2$,
cross-sectionally demeaned.

(2) Soft Spearman IC

Differentiable rank correlation via
pairwise sigmoids; temperature
anneals $0.20 \rightarrow 0.02$.

(3) Dispersion

$\kappa = \hat{\sigma}_y / \sigma_y \in [0.3, 3.0]$ —
penalises collapse and excessive
spread.

IC warmup: $\bar{\lambda}_{\text{ic}}$ ramps $0 \rightarrow \lambda_{\text{ic}}$ over 5 epochs so the model learns price representations before ranking pressure is applied.

Soft Spearman IC — Differentiable Ranking

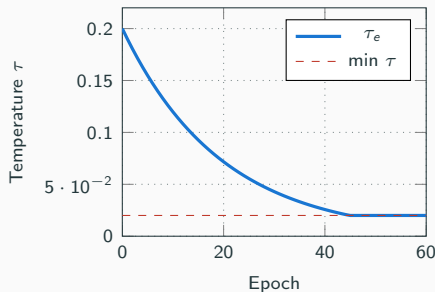
Standard Spearman is *non-differentiable*. Soft-rank via pairwise sigmoids:

$$r_i^{\text{soft}}(\hat{\mathbf{y}}) = \sum_{j=1}^N \sigma\left(\frac{\hat{y}_i - \hat{y}_j}{\tau_e}\right)$$

Temperature annealing

$$\tau_e = \max(0.02, 0.2 \cdot 0.95^e)$$

- **Early** (τ large): smooth gradients, exploration
- **Late** ($\tau \rightarrow 0.02$): near-discrete ranks, exploitation



Training Configuration — Three Variants

Hyperparameter	Base THGNN	Hybrid BiGRU	Mamba-MoE
Hidden / embed dim (d_h / D)	128	64	64
GAT heads	4	8	8
MAGE layers / MoE experts / MHA heads	—	1 / 4 / 2	1 / 4 / 2
Hyperedges (GPH / TCH)	—	32 / 32	32 / 32
Learning rate (peak)	1×10^{-4}	2×10^{-4}	1×10^{-4}
MSE / IC / Disp weights	0.7 / 0.5 / 0.6	0.4 / 0.8 / 0.1	0.6 / 0.6 / 0.1
IC warmup (epochs)	5	5	10
Trainable parameters	—	172,637	—
Time / epoch (RTX 3050 6 GB)	~ 23 s	~ 160 s	~ 200 s

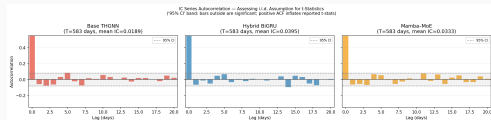
Optimizer AdamW, gradient clip $\|\nabla\|_2 \leq 1.0$, three-phase LR schedule (5-epoch warmup $\rightarrow$ 20-epoch plateau $\rightarrow$ cosine decay). **Splits:** train 2015–2022 | val 2023 | holdout 2024–2026.

Results

Quantitative Results — Holdout Rank-IC

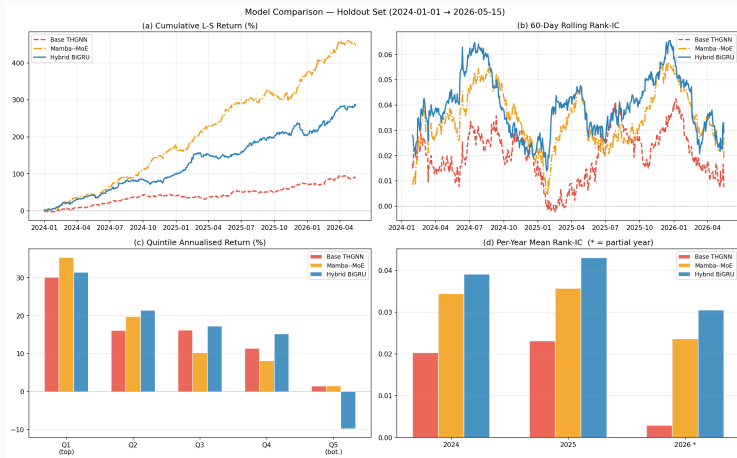
Model	Epoch	Rank-IC	MSE	Dir-Acc
Base THGNN	2	0.0204	1.3214	48.7%
Mamba-MoE	60	0.0343	1.2560	50.8%
Hybrid BiGRU	65	0.0398	1.1094	50.4%

- Hybrid BiGRU: +95.1% rank-IC over baseline, −16% MSE
- All three IC estimates statistically significant ($t \approx 9-17$, $p < 10^{-4}$)
- IC 0.02–0.04 matches published US/China results (MASTER: 0.025–0.060)



Daily IC autocorrelation — all lags within the 95% Bartlett band, validating the i.i.d. assumption behind the standard errors.

Holdout Backtest — Four Views

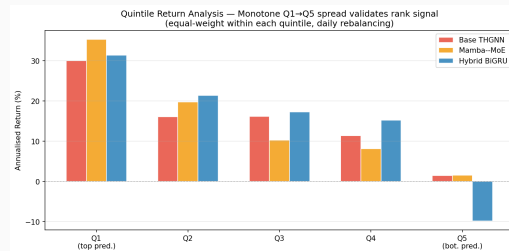


Equal-weight dollar-neutral long-short (top-5 – bottom-5), 583 trading days. Both hybrids **persistently outperform** the baseline; their 60-day rolling IC stays positive while the THGNN's collapses toward zero by 2026.

Backtest Summary & Quintile Spread

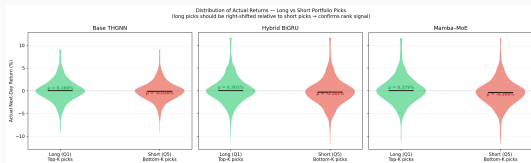
Model	IC	ICIR	Sharpe	MaxDD
Base THGNN	0.0189	0.206	2.30	−0.09
Mamba–MoE	0.0333	0.370	4.57	−0.09
Hybrid BiGRU	0.0395	0.420	3.71	−0.10

- Hybrid BiGRU: highest, **most consistent** signal (ICIR 0.420)
- Only the BiGRU reaches a **negative Q5 return** (−10.4%) — genuine *short-side* discrimination
- Mamba–MoE: best Sharpe via strongest short-leg mean

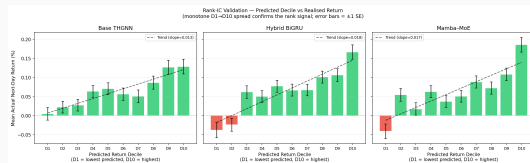


Monotone Q1→Q5 decline validates cross-sectional discriminative power across the full return distribution.

Return Distribution & Decile Validation

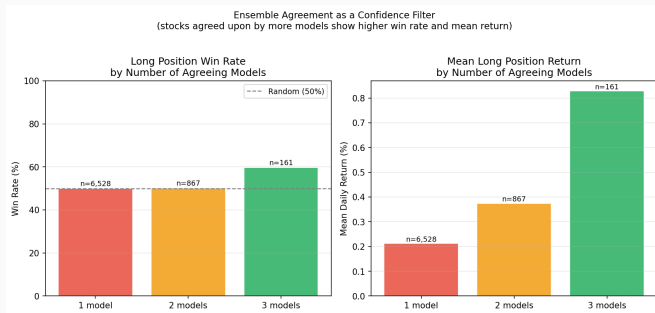


Long picks right-shifted vs. short. Mamba-MoE has the strongest short-leg mean (-0.309%) — its Sharpe premium.



Monotone decile spread; both hybrids yield negative bottom-decile returns (slopes: BiGRU 0.018, Mamba 0.017, THGNN 0.013).

Ensemble Agreement as a Confidence Filter

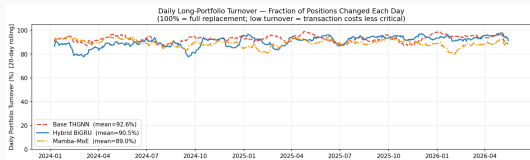


Consensus pays:

- 1 model picks: 50% win, +0.22%
- 2 models agree: +0.37%
- All 3 agree: 60% win, +0.83% — nearly 4× the single-model average

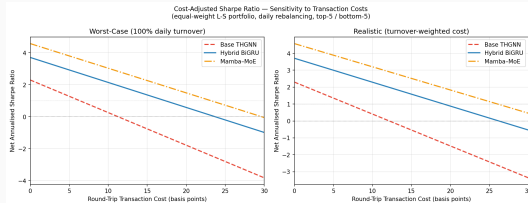
Variants capture **partially independent signals** — directly motivating the ensemble / stacking direction in future work.

Cost Realism — Turnover & Breakeven



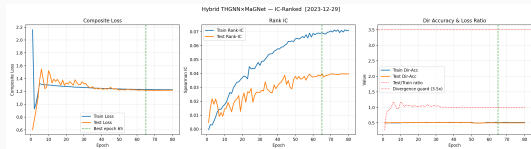
High turnover is structural: all models replace 89–93% of their top-5 book daily — near-full round-trip costs each day.

At a representative Indian round-trip cost of ~ 15 bp the baseline turns unprofitable, while **both hybrids retain positive net Sharpe**.



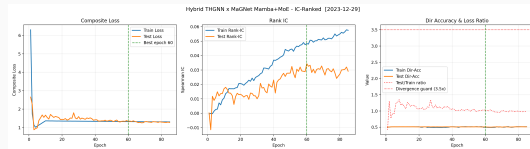
Sharpe breakeven costs: THGNN ≈ 10 bp, BiGRU ≈ 22 bp, **Mamba-MoE ≈ 28 bp** (most cost-robust).

Learning Dynamics — Hybrid Variants



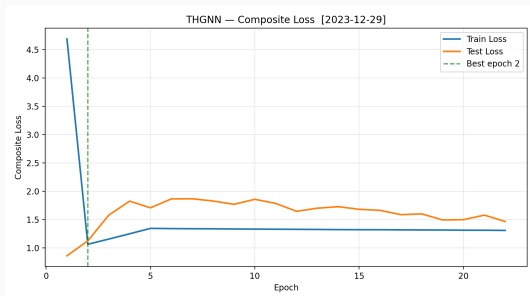
Hybrid BiGRU: healthy convergence; rank-IC grows monotonically to 0.040 at epoch 65, no overfitting.

Both hybrids exploit the **temperature-annealed soft-rank objective** to sharpen cross-sectional ranking as training matures.



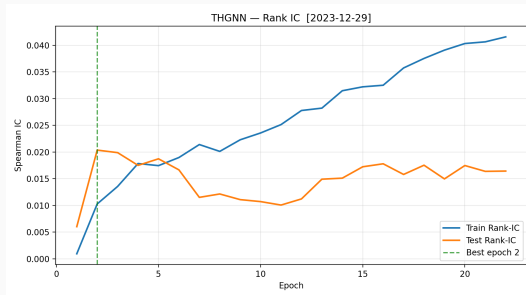
Mamba-MoE: validation loss plateaus from epoch 50; IC-patience termination at epoch 85.

Learning Dynamics — Baseline Saturation



Base THGNN: composite loss; early stopping triggers at epoch 22, best checkpoint saved at epoch 2.

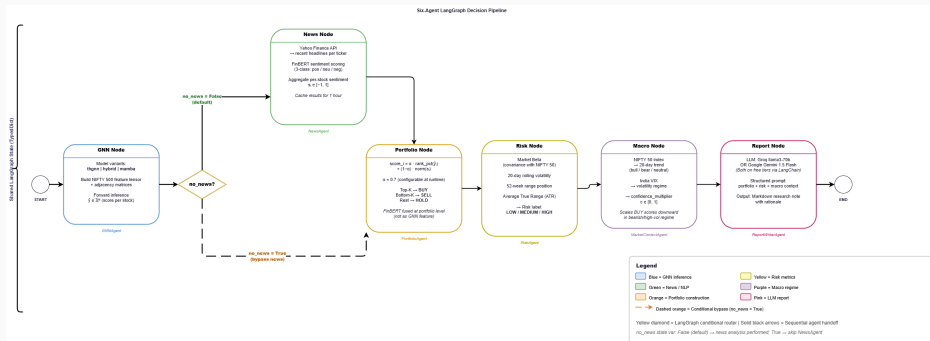
The simpler GRU-3-stream-GAT exhausts its representational capacity within a few gradient passes —
saturation, not instability.



Base THGNN rank-IC: saturates at epoch 2 then oscillates around 0.020 — the architecture's ceiling under its loss regime.

Multi-Agent Decision Pipeline

Six-Agent LangGraph Pipeline



- GNN** — trained checkpoint → score vector $\hat{y}$
- News** — FinBERT sentiment $s_i \in [-1, 1]$ per stock
- Portfolio** — fuse:

$$\text{score}_i = \alpha \text{rank}(\hat{y}_i) + (1-\alpha) \text{norm}(s_i)$$

Sentiment is fused at the **portfolio level** — α is tunable at inference without retraining.

- Risk** — beta, vol, 52-wk range, ATR → L/M/H label
- Macro** — NIFTY trend + India VIX → regime multiplier c
- Report** — Google gemma-4-31b → research note

Ticker	Action	GNN	Sent.	Fusion
JUBLPHARMA	BUY	1.000	—	1.000
EIDPARRY	BUY	0.900	—	0.900
DEEPAKNTR	BUY	0.950	+0.93	0.869
INDIGO	SELL	0.650	−0.07	0.446
RHIM	SELL	0.200	+0.93	0.420
ERIS	SELL	0.050	+0.94	0.331

- GNN **dominates** SELL leg even when sentiment is positive (RHIM: low rank → SELL)
- Negative sentiment correctly integrated (INDIGO — crude-oil headwinds)
- End-to-end latency ≈ 47 s (incl. LLM report)

Portfolio Recommendations — 2026-05-25

Model: Hybrid BiGRU (50 Top, $\alpha=0.60$) + News (Top 5000)

Buy Options

Market Regime: **BULL** | NIFTY up +1.82% | Idx trend: +1.75% | VIX: 16.7 | Conviction multiplier: 1.00x

BUY candidates: 10 | SELL candidates: 10 | Total stocks: 21

Full Ranking

Rank	Ticker	Action	gnn_score	sentiment_score	final_score	adjusted_score	rank_count
1	JUBLPHARMA	BUY	1.00000	0.00000	1.00000	1.00000	0
2	EIDPARRY	BUY	0.90000	0.00000	0.90000	0.90000	1
3	DEEPAKNTR	BUY	0.95000	0.93000	0.94000	0.94000	2
4	RHIM	SELL	0.20000	0.93000	0.42000	0.42000	3
5	AMINO	BUY	0.80000	0.00000	0.80000	0.80000	4
6	ERIS	SELL	0.05000	0.94000	0.33100	0.33100	5
7	INDIGO	SELL	0.65000	−0.07000	0.44600	0.44600	6
8	PHIL	BUY	0.70000	0.00000	0.70000	0.70000	7
9	CELESTIS	BUY	0.70000	0.00000	0.70000	0.70000	8
10	INDIGO	SELL	0.65000	−0.07000	0.44600	0.44600	9

Streamlit live portfolio tab: macro regime banner (BULL, VIX 16.7, conviction 1.00×) above the ranked allocation table.

Fusion-Weight Sensitivity (α Sweep)

α	GNN	Sent.	ρ vs GNN	SELL Δ
0.0	0%	100%	-0.094	4
0.3	30%	70%	+0.634	4
0.5	50%	50%	+0.795	3
0.7	70%	30%	+0.979	1
1.0	100%	0%	+1.000	0

Three findings:

1. BUY set is **entirely stable** from $\alpha=0.3$ to 1.0
2. SELL set is more sensitive — sentiment can *rescue* a borderline short with strong fundamental news
3. At $\alpha=0.7$, $\rho=0.979$ with pure GNN — predictions dominate, sentiment provides **targeted overrides**

Illustrative (single session, $N=21$); a multi-day sweep is future work.

Conclusion

Key Findings

What worked

- **Dual-hypergraph + MAGE** is the primary driver of the rank-IC gain (causal masking + latent group learning)
- **Composite IC loss** drives sustained improvement — IC grows to epoch 65 vs. saturation at epoch 2 for baseline
- **Ensemble consensus** carries 4× the expected return

Trade-offs

- **BiGRU** wins on rank-IC & ICIR $\Rightarrow$ best for *ranking* strategies
- **Mamba-MoE** wins on directional accuracy, Sharpe & cost-robustness $\Rightarrow$ best for *directional* signals
- Indian equity forecasting is *tractable but hard* — IC 0.02–0.04 is on par with US/China benchmarks

Limitations & Future Work

Limitations

- **Transaction costs**: 89–93% daily turnover; baseline unprofitable above ~ 15 bp
- Fixed temporal split — no walk-forward re-training
- Single-day horizon only
- **No component-level ablation** — the +95% gain blends architecture and loss-regime changes
- Inputs limited to price/volume; sentiment fused only at portfolio layer

Future work

1. **Loss-weight ablation** (highest priority) — grid / Bayesian search
2. Walk-forward backtest with monthly retraining
3. End-to-end sentiment fusion into the GNN tensor
4. **BiGRU + Mamba ensemble** — complementary IC & direction profiles
5. Multi-horizon prediction (1 / 5 / 20-day)
6. Cross-market generalisation (BSE 500, Nikkei, Hang Seng)

Thank you

Graph-Structured Deep Learning for Financial Forecasting

Hybrid THGNN–MaGNet | rank-IC (+95%) | NIFTY 500

Debarshi Chakraborty Questions?

References

-  Xiang et al. (2022). *Temporal and Heterogeneous Graph Neural Network for Financial Time Series Prediction*. CIKM.
-  MaGNet: Mamba Dual-Hypergraph Network for Stock Prediction.
-  Gu & Dao (2023). *Mamba: Linear-Time Sequence Modeling with Selective State Spaces*.
-  Li et al. (2024). *MASTER: Market-Guided Stock Transformer*. AAAI.
-  Sawhney et al. (2021). *Stock Selection via Spatiotemporal Hypergraph Attention Network*. AAAI.
-  Zhao & Akoglu (2020). *PairNorm: Tackling Oversmoothing in GNNs*. ICLR.
-  Araci (2019). *FinBERT: Financial Sentiment Analysis with Pre-trained Language Models*.